Classification of Semisimple Lie Algebras

Mathematical Physics – Differential Geometric Methods

Master's Degree in Mathematics

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UNIVERSITÀ DI TRENTO

This seminar is meant to be an introduction to the classification of semisimple Lie algebras through root systems and Dynkin diagrams, following sections of Parts II and III of *Introduction to Lie algebras and representation theory* by James E. Humphreys (Springer, 1972).

Since we don't have the time to develop the entire theory we need, some technical results will be stated without proof: this way we will be able to analyse more in detail the theorem giving the isomorphism between two Lie algebras with the same root system.

Notes for this seminar are available at
`davideborra.github.io/?r=Lie`

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1 Introduction

- Introduction
- Root systems and their classification
- Root space decomposition
- Isomorphism Theorem

Definition (Lie algebra).

A *Lie algebra* over a field $\mathbb{K}$ is a $\mathbb{K}$-vector space L endowed with an operation $[\cdot,\cdot] : L \times L \to L$, called *Lie bracket* or *commutator* such that

(1) the bracket is bilinear;

(2) $[x, x] = 0$ for any $x \in L$

(3) $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$ for any $x, y, z \in L$ (*Jacobi identity*).

Examples. $\mathfrak{gl}(V) := \operatorname{End} V$, $\mathfrak{sl}(V) := \{f \in \mathfrak{gl}(V) \mid \operatorname{tr} f = 0\}$, where V is a vector space over a field $\mathbb{K}$, with $[f, g] := f \circ g - g \circ f$.

Adjoint representation

Any Lie algebra comes with a map ad, called "adjoint representation" and defined as follows:

$$\begin{array}{rccccl} \mathrm{ad}: & L & \longrightarrow & \mathrm{Der}\, L & & \\ & x & \longmapsto & \mathrm{ad}\, x: & L & \longrightarrow & L \\ & & & & y & \longmapsto & \mathrm{ad}\, x\, y = [y, x]. \end{array}$$

A Lie algebra is called Abelian if ad is the zero map.

More introductory definitions

- $I \subseteq L$ is called an ideal if $[x, y] \in I$ for any $x \in I, y \in L$;

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- Given any $X \subseteq L$, the *centraliser* of X in L is $C_L(X) := \{y \in L \mid [y, X] = 0\}$;
- The derived series of a Lie algebra is defined inductively by setting $L^{(0)} := L$ and $L^{(n+1)} := [L^{(n)}, L^{(n)}]$;

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- A lie algebra is called semisimple if $\mathrm{Rad}\, L = 0$.

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2 Root systems and their classification

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Framework

Throughout this section, we will need A real vector space V, endowed with a scalar product $(\cdot,\cdot)$, and we will be calling (for $\alpha \in H$)

- $H_\alpha := \{\beta \in V \mid (\alpha,\beta) = 0\}$ the hyperplane orthogonal to α;
- σ_α the reflection with respect to H_α

$$\sigma_\alpha(\beta) = \beta - \frac{2(\alpha,\beta)}{(\alpha,\alpha)}\alpha, \qquad \beta \in V.$$

- $\langle \beta, \alpha \rangle = \beta - 2\frac{(\alpha,\beta)}{(\alpha,\alpha)}$

Definition (Root systems).

A *root system* of rank ℓ in V is a subset Φ of an ℓ-dimensional vector space V such that

(1) Φ is finite, spans V and does not contain 0;

(2) if $\alpha \in \Phi$, the only multiples of α in Φ are $\pm\alpha$;

(3) if $\alpha \in \Phi$, the reflection σ_α leaves Φ invariant.

(4) if $\alpha, \beta \in \Phi$, then $\langle \beta, \alpha \rangle \in \mathbb{Z}$.

Φ is called *reducible* if it can be partitioned into the union of two proper subsets such that each root in one subset is orthogonal to each root in the other, and *irreducible* otherwise.

Weyl group

Weyl group

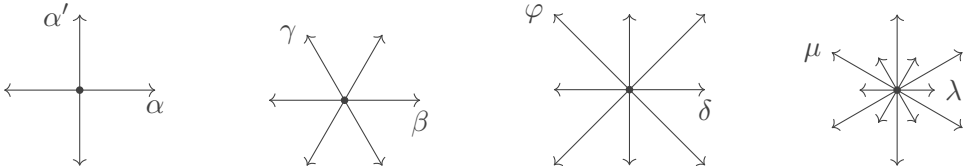
The *Weyl group* of a root system Φ is the subgroup of $\mathrm{GL}(V)$ generated by the reflections σ_α, for $\alpha \in \Phi$:

$$\mathscr{W} = \langle \{\sigma_\alpha \mid \alpha \in \Phi\} \rangle.$$

Examples

If $\ell = 1$, there is only one root system:

If $\ell = 2$, things get more interesting:



$A_1 \times A_1$ A_2 B_2 G_2

Figure: Some root systems in $\mathbb{R}^2$

Angles

Condition (4) puts a strong limit on which angles ϑ can there be between any two roots α and β in a root system Φ. By definition of cosine

$$4\cos^2\vartheta = \langle\alpha,\beta\rangle\langle\beta,\alpha\rangle \in \mathbb{Z},$$

hence $\cos\vartheta$ can be either 0 or $\pm\frac{1}{2}$, forcing ϑ to assume the values in the table.
It will be key later that the product $\langle\alpha,\beta\rangle\langle\beta,\alpha\rangle$ can assume only values 0,1,2 or 3.

$\langle\alpha,\beta\rangle$	$\langle\beta,\alpha\rangle$	ϑ
0	0	$\pi/2$
1	1	$\pi/3$
-1	-1	$2\pi/3$
1	2	$\pi/4$
-1	-2	$3\pi/4$
1	3	$\pi/6$
-1	-3	$5\pi/6$

Bases

Definition (Base).

A subset $\Delta \subset \Phi$ is called a *base* of the root system Φ if

(1) Δ is a basis of V;

(2) each $\beta \in \Phi$ can be written as a linear combination $\beta = \sum_{\alpha \in \Delta} b_\alpha \alpha$, where the b_α's are all nonnegative or nonpositive.

The roots in a base are called *simple*.

One may prove bases always exist. Moreover, that the angle between two simple roots α and β is always obtuse, hence the values $\langle \alpha, \beta \rangle$ and $\langle \beta, \alpha \rangle$ will always be nonpositive. Similarly, $\alpha - \beta$ cannot be a root.

Ordering induced by a base

Each base Δ of a root system Φ comes with an induced ordering on Φ: for any β, γ,

$$\beta \prec_\Delta \gamma :\iff \gamma - \beta = \sum_{\alpha \in \Delta} k_\alpha \alpha \text{ and } k_\alpha \geq 0 \; \forall \alpha \in \Delta.$$

Lemma

Let Φ be an irreducible root system and Δ a base. Then there exists a unique maximal root with respect to $\prec_\Delta$.

Cartan matrices

Definition (Cartan matrix)

Φ root system, $\Delta = \{\alpha_1, \ldots, \alpha_\ell\}$ base. The Cartan matrix of a root system is the matrix

$$\big(\langle \alpha_i, \alpha_j \rangle\big)_{ij=1}^{\ell}.$$

For example, here we compute the Cartan matrices of the root systems of rank 2 are

$$\mathsf{A}_1 \times \mathsf{A}_1 : \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}; \quad \mathsf{A}_2 : \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}; \quad \mathsf{B}_2 : \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}; \quad \mathsf{G}_2 : \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$$

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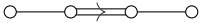
1. draw ℓ vertexes, each corresponding to a root α_i;
2. connect the i-th vertex and the j-th vertex with $\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle$ edges;
3. draw an arrow pointing towards the vertex corresponding to the shortrest root.

Example: F_4

Consider the following Dynkin diagram, associated to the root system Φ of type F_4.



We compute its Cartan matrix.

$$\begin{pmatrix} \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \end{pmatrix}$$

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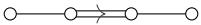

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$$\langle \alpha_1, \alpha_2 \rangle \langle \alpha_2, \alpha_1 \rangle = 1;$$

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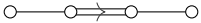

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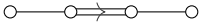

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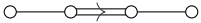

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$$\langle \alpha_i, \alpha_i \rangle \langle \alpha_i, \alpha_i \rangle = 2;$$

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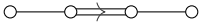

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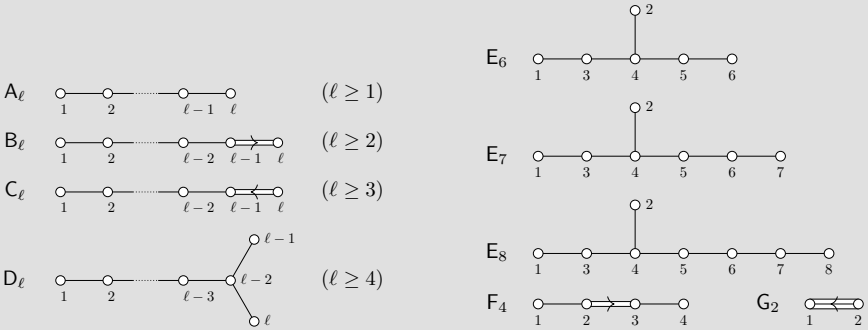
$$\langle \alpha_i, \alpha_i \rangle \langle \alpha_i, \alpha_i \rangle = 2;$$
$$\langle \alpha_i, \alpha_j \rangle \langle \alpha_j, \alpha_i \rangle = 0 \text{ otherwise;}$$

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -2 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

Classification of irreducible root systems

Theorem

If Φ is an irreducible root system of rank ℓ, then its Dynkin diagram is one of the following.



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Toral subalgebras

Any Lie algebra can be interpreted as a subalgebra of $\mathfrak{gl}(V)$ for some vector space V. This allows us to perform Jordan–Chevalley decomposition, writing $x \in L$ as

$$x = x_s + x_n.$$

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- an $x \in L$ is called **semisimple** if the roots of its minimal polynomial are all distinct or, equivalently (if $\mathbb{K}$ is algebraically closed), if it is diagonalisable.
- a subalgebra is called **toral** if it contains only semisimple elements.

Eigenspace decomposition

3 Root space decomposition

Any Li algebra L contains a maximal toral subalgebra H, which is Abelian. Then $\operatorname{ad} H$ is a commuting family of commuting semisimple (diagonalisable) endomorphisms: therefore a basis of common eigenvectors exists.

$$L = L_0 \oplus \bigoplus_{\alpha \in \Phi} L_\alpha$$

$$L_\alpha = \{x \in L \mid \operatorname{ad} h(x) = \alpha(h)x \; \forall h \in H\}, \qquad \Phi = \{\alpha \in H^\vee \mid L_\alpha \neq 0\}$$

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Killing form

Definition (Killing form).

Let L be any Lie algebra. The Killing form on L is the map $\kappa : L \times L \to \mathbb{K}$ given by

$$\kappa(x, y) = \operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y) \qquad \text{for } x, y \in L.$$

The Killing form is a symmetric bilinear form, and it satisfies

$$\kappa([x, y], z) = \kappa(x, [y, z]).$$

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Theorem

Let L be a Lie algebra, then L is semisimple if and only if its Killing form is nondegenerate.

Killing dual

H can be identified with $H^\vee$ via an isomorphism given by the Killing form: to a $\phi \in H^\vee$ corresponds the (unique) element $t_\phi \in H$ satisfying

$$\phi(h) = \kappa(t_\phi, h) \qquad \forall h \in H.$$

This allows us to pull back the killing form to $H^\vee$, defining a new bilinear form

$$\kappa^\vee(\phi, \psi) := \kappa(t_\phi, t_\psi).$$

Furthermore, the restriction of the Killing form to H is nondegenerate.

Orthogonality

3 Root space decomposition

Proposition (Orthogonality /1).

(1) Φ spans $H^\vee$;

(2) $\kappa(L_\alpha, L_\beta) = 0$ for any $\alpha, \beta \in \Phi$ such that $\alpha + \beta \neq 0$.

(3) if $\alpha \in \Phi$, then $-\alpha \in \Phi$;

(4) let $\alpha \in \Phi$, $x \in L_\alpha$, $y \in L_{-\alpha}$. Then $[x, y] = \kappa(x, y)t_\alpha$

(5) if $\alpha \in \Phi$, $[L_\alpha, L_{-\alpha}] = \mathbb{K}t_\alpha \leq_{\mathbb{K}} H$ is one-dimensional;

(6) $\alpha(t_\alpha) = \kappa(t_\alpha, t_\alpha) \neq 0$, for $\alpha \in \Phi$

Orthogonality

3 Root space decomposition

Proposition (Orthogonality /2).

(1) If $\alpha \in \Phi$ and x_α is any nonzero element of L_α, then there exists $y_\alpha \in L_{-\alpha}$ such that x_α, y_α and $h_\alpha = [x_\alpha, y_\alpha]$ span a three-dimensional simple subalgebra of L isomorphic to $\mathfrak{sl}(2, \mathbb{K})$ via

$$x_\alpha \mapsto \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad y_\alpha \mapsto \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad h_\alpha \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix};$$

(2) $h_\alpha = \frac{2t_\alpha}{\kappa(t_\alpha, t_\alpha)}$; $h_\alpha = -h_{-\alpha}$, $\alpha(h_\alpha) = 2, h_\alpha \in H$.

Rationality /1

We want to produce a root system, so we need a real vector space.

- First, we observe that every root in Φ can be expressed as a $\mathbb{Q}$-linear combination of roots in a base Δ;
- Consider $V_{\mathbb{Q}} := \operatorname{span}_{\mathbb{Q}} \Phi \subset H^{\vee}$: $\dim_{\mathbb{Q}} V_{\mathbb{Q}} = \ell$
- $\kappa^{\vee}_{|V_{\mathbb{Q}} \times V_{\mathbb{Q}}}$ is a positive-definite symmetric bilinear form.
- Extend scalars $V := V_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{R}$ and call $(\cdot, \cdot) = \kappa^{\vee}_{|V_{\mathbb{Q}} \times V_{\mathbb{Q}}} \otimes \mathrm{Id}_{\mathbb{R} \times \mathbb{R}}$

Rationality /2

Theorem (Rationality).

L, H, Φ, V as above. Then

(1) Φ spans V, and 0 does not belong to Φ;

(2) if $\alpha \in \Phi$, then $-\alpha \in \Phi$ and no other scalar multiple of Φ is a root.

(3) if $\alpha, \beta \in \Phi$, then $\beta - \frac{2(\alpha,\beta)}{(\alpha,\alpha)}\alpha \in \Phi$;

(4) if $\alpha, \beta \in \Phi$, then $\frac{2(\alpha,\beta)}{(\alpha,\alpha)} \in \mathbb{Z}$;

i.e. Φ is a root system.

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Reduction to the simple case

4 Isomorphism Theorem

We will be discussing slightly weaker results concerning simple Lie algebras, as the generalisation to the semisimple case is straightforward.

Proposition

Let L be a simple Lie algebra, H a maximal toral subalgebra of L and $\Phi \subset H^\vee$ the set of roots of L relative to H. Then Φ is an irreducible root system.

Proposition

Let L be a semisimple Lie algebra, H a maximal toral subalgebra of L and $\Phi \subset H^\vee$ the set of roots of L relative to H. Fix a base Δ of Φ: then L is generated (as a Lie algebra) by the root spaces L_α, $L_{-\alpha}$ $(\alpha \in \Delta)$ or, equivalently, by root vectors $x_\alpha \in L_\alpha$ and $y_\alpha \in L_{-\alpha}$ $(\alpha \in \Delta)$

Statement of the Theorem

Theorem

Let L, L' be simple Lie algebras over $\mathbb{K}$, with respecive maximal toral subalgebras H, H' and corresponding root systems Φ, Φ'. Suppose there is an isomorphism of Φ onto Φ' (denoted $\alpha \mapsto \alpha'$), inducing $\pi : H \to H'$. Fix a base $\Delta \subset \Phi$, so $\Delta' = \{\alpha' \mid \alpha \in \Delta\}$ is a base of Δ'. For each α choose arbitrary algebra isomorphisms $\pi_\alpha : L_\alpha \to L_{\alpha'}$. Then there exists a unique isomorphism $\bar{\pi} : L \to L'$ extending $\pi : H \to H'$ and all the π_α's ($\alpha \in \Delta$).

Sketch of the proof /1: framework

Uniqueness follows directly by the previous results. We focus on existence.

- We exploit the fact that if L and L' are essentially the same, then their direct sum $L \oplus L'$ contains a subalgebra (the "diagonal") isomorphic to L by the projection on either factor.

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- We exploit the fact that if L and L' are essentially the same, then their direct sum $L \oplus L'$ contains a subalgebra (the "diagonal") isomorphic to L by the projection on either factor.
- Call D the subalgebra generated by

$$\tilde{x}_\alpha = (x_\alpha, x'_{\alpha'}), \qquad \tilde{y}_\alpha = (y_\alpha, y'_{\alpha'}), \qquad \tilde{h}_\alpha = (h_\alpha, h'_{\alpha'}) \qquad \forall \alpha \in \Delta.$$

Sketch of the proof /2: $D \subsetneq L \oplus L'$

- We prove D is properly contained in $L \oplus L'$

Sketch of the proof /2: $D \subsetneq L \oplus L'$

- We prove D is properly contained in $L \oplus L'$
  - We prove $\operatorname{ad} D$ stabilises M, where M is spanned by the

$$A_{\alpha_1,\ldots,\alpha_k}(\tilde{x}) = \operatorname{ad} \tilde{\gamma}_{\alpha_1} \operatorname{ad} \tilde{\gamma}_{\alpha_1} \cdots \operatorname{ad} \tilde{\gamma}_{\alpha_k} \tilde{x} \qquad (\alpha_i \in \Delta \text{allowing repetitions}).$$

where we chose nonzero $x \in L_\beta$ and $x' \in L'_{\beta'}$, set $\tilde{x} := (x, x') \in L \oplus L'$, for β maximal in Φ.

Sketch of the proof /2: $D \subsetneq L \oplus L'$

- We prove D is properly contained in $L \oplus L'$
  - We prove $\operatorname{ad} D$ stabilises M, where M is spanned by the

$$A_{\alpha_1,\ldots,\alpha_k}(\tilde{x}) = \operatorname{ad}\tilde{\gamma}_{\alpha_1}\operatorname{ad}\tilde{\gamma}_{\alpha_1}\cdots\operatorname{ad}\tilde{\gamma}_{\alpha_k}\,\tilde{x} \qquad (\alpha_i \in \Delta \text{allowing repetitions}).$$

    where we chose nonzero $x \in L_\beta$ and $x' \in L'_{\beta'}$, set $\tilde{x} := (x, x') \in L \oplus L'$, for β maximal in Φ.
  - This proves D must be proper: otherwise M would be a proper nonzero ideal of $L \oplus L'$ but, L and L' being simple, they are the only ideals of this type.

Sketch of the proof /3: conclusion

- The projections are injective as otherwise this forces $D = L \oplus L'$, and surjective by construction and the previous Proposition,

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$\square$

Conclusions

4 Isomorphism Theorem

$$\left\{\text{Simple Lie algebras}\right\} \rightleftarrows \left\{\text{Irreducible root system}\right\} \rightleftarrows \left\{\text{Dynkin diagrams}\right\}$$

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$$\left\{\text{Semisimple Lie algebras}\right\} \rightleftarrows \left\{\text{Root system}\right\} \rightleftarrows \left\{\begin{array}{c}\text{Products of}\\ \text{Dynkin diagrams}\end{array}\right\}.$$

Classification of Semisimple Lie Algebras

Thank you for listening!
Any questions?